# Manual

## MES-Cockpit Client Shop Floor Information

## MC-CSI 3.1

Version 1.0.23049

Last changed on: 01.09.2020

## Copyright

# Contents

# 1 MES-Cockpit Client Shop Floor Information

## Purpose

The function package Client Shop Floor Information in MES-Cockpit provides extensive functions to display instance-related online information on the following objects:

- Workplaces/machines
  Tabular overview of workplaces/machines to visualize the current machine condition (status, point in time since when the status is applicable) including basic information on the workplace/machine.
- KPI monitor
  When starting the KPI monitor, it first provides an overview of the whole site and enables to drill down key figures in relation to cost centers, machine groups and the single object "machine".
- Messages listing
  List of planned maintenance orders (pool of orders including orders/OPs of the "maintenance" category).
- Contact person
  Search for contact partners within the company and check if they are available.

# 2 Workplaces/Machines

## 2.1 General

| | |
|---|---|
| App name | Workplaces/Machines |
| Short name of app | Workplaces |
| Function authorization | sma.wpov |

The tabular overview *Workplaces/Machines* visualizes the current machine status (status, point in time since when the status is available) including basic information on the workplace/machine. The application helps to increase transparency in the shop floor. The user gets a clear overview of the current statuses of machines and workplaces.

### Selection criteria

The application provides the following selection criteria:

**Resource**

Enter the resource number (workplace number) to select from the workplaces.

**Status text**

Select the relevant status text to select the status that is displayed.

### Field descriptions – List

**Image**

Shows the picture of the machine stored in the configuration of workplaces and resources.

Note: The picture is loaded when the selection is made. Before selection, a dummy image is shown for the different machines.

**Workplace number - workplace designation**

Shows the workplace number and workplace name. The system only shows the workplaces that are included in the responsibility area the user is authorized for.

**Group**

Workplace group of the machine

**Status**

The respective workplace statuses are presented with the following colors used for the available resource performance accounts:

- RPA 1: dark-green

- RPA 2: red
- RPA 3: pink (fuchsia)
- RPA 4: purple
- RPA 5: black
- RPA 6: dark-gray
- RPA 7: light turquoise
- RPA 8: pale blue
- RPA 9: dark blue
- RPA 10: brown
- RPA 11: light green
- RPA 12: turquoise

## Field descriptions – Detail

**Image**

See description "List".

**Workplace number - workplace designation**

See description "List".

**Status**

See description "List".

**Status text**

Defined status text of the current status

**Status since**

Date when the status was set

**Duration so far**

Duration of the status (until now)

**Expected end**

If specified, the estimated end of the current status

**Article**

Article logged on to the machine

**Article name**

Name of the article/item logged on to the machine

**MES order number**

MES order number of the operation currently logged on. If several operations are logged on to the workplace at the same time, the operation is displayed that was logged on last.

If you click the MES order number, you are redirected to the *Operation overview*.

**OP name**

Name of the OP logged on

**Company**

Company of the workplace

**Cost center**

Cost center of the workplace.

**Group**

Workplace group of the workplace

## 2.2 Functions

Subject to the purchased licenses and the available authorizations, the application *Workplaces/Machines* provides the following functions:

**TOP 5 of malfunctions during shift**

Shows the ranking list of the TOP 5 downtimes at the machine including information on the duration and how frequently the malfunction occurred at the respective machine.
It is also possible to show the TOP 5 based on the frequency or total duration.

**Running OPs (Operations logged on)**

The list shows all operations logged on to the machine. If you click the operation, you can change to the *Operation overview.*

**Last 10 malfunctions**

The list shows the 10 malfunctions that last occurred at the machine.

**Change status**

Use this function to change the status of the selected machine. A list of the possible machine statuses is provided. Select the relevant status.
**Note**: This function is included in the package SMA-AMF. Function authorization: sma.setstatus

**Operation overview**

Click this button to switch to the *Operation overview* and to display detail information on the operation, which was displayed in the detail view of the machine.

**Log operation on**

Use the dialog *Log operation on* to log an operation on to the currently selected workplace. Select the operation (MES order number) and machine status via search dialog from a list.

**Note:** All HYDRA standard validation checks are performed.

**Note**: This function is included in the package SMA-AMF. Function authorization: sma.logon

**Log operation off**

Use the dialog *Log operation off* to log an operation off from the currently selected workplace. You can use the logoff dialog to enter the operation number (MES order number), yield and scrap quantities, the relevant scrap reason and a new machine status. You can use a search dialog to select the MES order number, the scrap reason and the machine status from a list.

**Note:** All HYDRA standard validation checks are performed.

**Note**: This function is included in the package SMA-AMF. Function authorization: sma.logoff

**Interrupt operation**

Use the dialog *Interrupt operation* to interrupt an operation logged on to the currently selected workplace.

You can enter the operation number (MES order number), yield and scrap quantities, the relevant scrap reason and a new machine status in the dialog. You can use a search dialog to select the MES order number, the scrap reason and the machine status from a list.

**Note:** All HYDRA standard validation checks are performed.

**Note**: This function is included in the package SMA-AMF. Function authorization: sma.interrupt

**Posting of part quantity (partial confirmation)**

You can use the *Partial confirmation* dialog to post part quantities for an operation logged on to the currently selected workplace.

You can enter the operation number (MES order number), yield and scrap quantities and the relevant scrap reason in the dialog. You can use a search dialog to select the MES order number and the scrap reason.

**Note:** All HYDRA standard validation checks are performed.

**Note**: This function is included in the package SMA-AMF. Function authorization: sma.partial conf

**Change partitioning**

Use the *Change partitioning* dialog to change the current partitioning for the workplace currently selected.

The operation number (MES order number) and the new partitioning can be entered in the dialog. You can use a search dialog to select the MES order number from the list of operations logged on.

**Note:** All HYDRA standard validation checks are performed.

**Note**: This function is included in the package SMA-AMF. Function authorization: sma.partition

## Application settings

In this dialog, you can make specific settings for this application.

- *Reload machine status:*
  - No: The open application is not automatically updated.
  - Cyclic loading: If this option is set, the status of the displayed workplaces/machines is cyclically updated. The duration between two updates is defined via the option *Reload time*.
  - EMQTT: The machine status is promptly updated when the status of the machine changes (via EMQTT). Requirement: You use the centralized MDE and the relevant settings are made. If data collection is not performed via centralized MDE for the machine and if the machine is then not updated via MQTT, an update in the application *Workplaces/Machines* is also not possible.
- *Reload time:* The reload time specifies the time between the status updates. You can define a value between 30 and 300 seconds as reload time.

To enable the settings, click *Save* after having performed the changes and close the application using .

# 3 KPI Monitor

## 3.1 General

| | |
|---|---|
| App name | KPI monitor |
| Short name of app | KPI |
| Function authorization | sma.kpim |

When you open the application *KPI monitor,* you get a clear an overview of the KPIs defined for the factory/site. You can also get more detailed information and drill down the different KPIs in relation to the integrated dimensions and the relevant object "machine".

The *KPI monitor* currently provides the following KPIs:

- OEE and its components
  The *KPI monitor* (start screen) shows the OEE and its components for all machines (site) as a bar chart in the *OEE* section.
  - OEE
  - OEE performance
  - OEE availability
  - OEE quality

  The calculation is based on the formulas defined in HYDRA.
  If you click the OEE section, you are forwarded to the detail application OEE.
- Rate of capacity utilization (utilization efficiency)
  The KPI monitor (start screen) shows the current KPI rate of capacity utilization for all machines (site) in a horizontal bar chart in the *Rate of capacity utilization* section.
  If you click the *Rate of capacity utilization* section, you are forwarded to the detail application *Rate of capacity utilization.*
- Produced quantities
  The KPI monitor (start screen) shows the produced yield and scrap quantity for all machines (site) in a horizontal bar chart in the *Produced quantities* section.
  If you click the *produced quantities* section, you are forwarded to the detail application *produced quantities*.
- Scrap rate
  The KPI monitor (start screen) shows the scrap rate for all machines (site) in a bar chart in the *scrap rate* section.
  If you click the *scrap rate* section, you are forwarded to the detail application *scrap rate.*

- Production times and downtimes
  The KPI monitor (start screen) shows the production times and downtimes for all machines (site) in a in a pie chart in the *production and downtimes* section.
  If you click the *production and downtimes* section, you are forwarded to the detail application *production times and downtimes.*
- Failure report
  The KPI monitor (start screen) shows the produced yield and failure (defective) quantity for all machines (site) in a horizontal bar chart in the *failure report* section. The failure quantity results from the number of recorded failure types and their weighting.
  If you click the *failure report* section, you are forwarded to the detail application.

## 3.2 Detail application *OEE*

Graphic presentation (diagram) of the KPI *OEE* and its components for the object selected in the list.

Please note: The list only shows the workplaces for which data is available to calculate the OEE.

#### *3.2.1.1 Functions of the detail application*

- Changing the display layout
  You can switch between the following display options:
  - Horizontal bar chart
  - Vertical bar chart
- Changing the dimension displayed
  You can display KPIs for the following dimensions in the detail view:
  - Workplace group/machine group
  - Cost center
  - Workplace/machine

## 3.3 Detail application *Rate of capacity utilization*

Graphic presentation (diagram) of the KPI *rate of capacity utilization* (green) for the object selected in the list.
The KPI *rate of capacity utilization* is calculated as follows:

$$Rate\ of\ capacity\ utilization = \frac{RPA_{11}}{\sum RPA_{1-11}}$$

Please note: The list only shows the workplaces for which data is available to calculate the *rate of capacity utilization.*

#### 3.3.1.1 Functions of the detail application

- Changing the display layout
  You can switch between the following display options:
  - Horizontal bar chart
  - Vertical bar chart
  - Pie chart
- Changing the dimension displayed
  You can display KPIs for the following dimensions in the detail view:
  - Workplace group/machine group
  - Cost center
  - Workplace/machine

## 3.4 Detail application *Produced quantities*

Graphic presentation of the produced yield and scrap quantities each in the "primary" quantity unit.

Please note: The list only shows the workplaces for which data is available to calculate the produced quantities.

#### 3.4.1.1 Functions of the detail application

- Changing the display layout
  You can switch between the following display options:
  - Horizontal bar chart
  - Vertical bar chart
- Changing the dimension displayed
  You can display KPIs for the following dimensions in the detail view:
  - Workplace group/machine group
  - Cost center
  - Workplace/machine

## 3.5 Detail application *Scrap rate*

Graphic presentation (diagram) of the KPI *scrap rate* for the object selected in the list. The KPI *scrap rate* is calculated as follows:

$$Scrap\ rate = \frac{Scrap_{Primary}}{Scrap_{Primary} + Yield_{Primary}}$$

Please note: The list only shows the workplaces for which data is available to calculate the scrap rate.

#### 3.5.1.1 Functions of the detail application

- Changing the display layout
  You can switch between the following display options:
  - Horizontal bar chart
  - Vertical bar chart
  - Pie chart
- Changing the dimension displayed
  You can display KPIs for the following dimensions in the detail view:
  - Workplace group/machine group
  - Cost center
  - Workplace/machine

## 3.6 Detail application *Production times and downtimes*

Graphic presentation (diagram) of production times and downtimes for the object selected in the list. Production time refers to the time posted to the resource performance account (RPA) 11 and downtime refers to the durations posted to the resource performance accounts (RPA) 1-10.

Please note: The list only shows the workplaces for which data is available to calculate production times and downtimes.

#### 3.6.1.1 Functions of the detail application

- Changing the display layout
  You can switch between the following display options:
  - Horizontal bar chart
  - Vertical bar chart
- Changing the dimension displayed
  You can display KPIs for the following dimensions in the detail view:
  - Workplace group/machine group
  - Cost center
  - Workplace/machine

## 3.7 Detail application *Failure report*

Graphic presentation of the produced yield in the "primary" quantity unit and the failure (defective) quantity of quality inspections. The failure quantity results from the number of recorded failure types and their weighting.

Please note: The list only shows the workplaces for which data is available to calculate the produced quantities.

#### 3.7.1.1 Functions of the detail application

- Changing the display layout
  You can switch between the following display options:
  - Horizontal bar chart
  - Vertical bar chart
- Changing the dimension displayed
  You can display the KPI *failure rate in ppm* for the following dimensions in the detail view:
  - Workplace group/machine group
  - Cost center
  - Workplace/machine

The KPI *failure rate in ppm* is calculated as follows:

$$Scrap\ rate = \frac{Failure\ quantity}{Scrap_{Primary} + Yield_{Primary}} * 1.000.000$$

## 3.8 General – KPI Monitor

#### 3.8.1.1 Selection/filter options of the KPI monitor

The *KPI monitor* provides the following selection options:

**Date from / to**

The selection criterion *Date from/to* specifies the period of time you want to evaluate.

**Filter mode time/shift**

Use the *filter mode* to specify whether the selection is based on a time or shift specified.

**Shift**

If you use the selection criterion *Shift,* you select shifts to specify the evaluated period of time (date from ... to ...). If the selection period exceeds the period for the online data area, the system automatically selects the medium-term data area.

**Time from/to**

If you use the selection criterion *Time,* you select a time (from... to...) to specify the evaluated period of time (date from ... to ...). If the selection period exceeds the period for the online data area, the system automatically selects the medium-term data area.

When you open the KPI monitor, the system uses the default selection for the display, which is defined in the application settings. If you change the selection, this selection applies for the duration of the session.

Use the button *Reset* to reset the selection to the default selection defined in the application settings.

If you click a detail application, the selection criteria are used for the detail application.

The *Failure report* cannot show any evaluations based on shifts. If you have made a selection by shift, this report does not display any key figure then. The KPIs in the *Failure report* are only available if you have made a selection by *Time from/to*.

### 3.8.1.2 Application settings in the KPI monitor

The *KPI monitor* provides the following application settings. These settings are used to predefine the selection criteria used to open the *KPI monitor:*

**Default date in filter**

Use the field *Default date in filter* to assign the current date (date when KPI monitor is called) to the selection field *Date from/to* or a calculated date with a relative specification. If you specify a relative date, the following options are available:

- Number with the signs + and -
- Time period with the following options: days/weeks/months/years

Example:

| Date of call | Default date in filter (relative) | Transfer to the selection field *Date from/to* |
|---|---|---|
| **15.05.2019** | from: -2 days<br>to: + 3 days | from: 13.05.2019<br>to: 18.05.2019 |

**Preset filter mode**

If you use the field *Preset filter mode*, the selection field *filter mode* is predefined, which specifies whether the selection is based on a time or shift specified.

**Default shift**

Use the field *Default shift* to predefine the selection field *Shift,* which specifies the evaluated period of time (date from ... to ...) using the shifts selected.

**Time from/to**

Use the field *Time from/to* to predefine the selection field *Time from/to,* which specifies the evaluated period of time (date from ... to ...) using the time (from... to...) selected.

# 3.9 General – Detail applications

### *3.9.1.1 Selection/filter options in the detail applications*

The detail applications provide the following selection options:

**Resource**

Enter the resource number (workplace number) to select from the workplaces.

**Group**

Workplace group

**Cost center**

Cost center of workplaces

**Date from / to**

The selection criterion *Date from/to* specifies the period of time you want to evaluate.

**Filter mode time/shift**

Use the *filter mode* to specify whether the selection is based on a time or shift specified.

**Shift**

If you use the selection criterion *Shift,* you select shifts to specify the evaluated period of time (date from ... to ...). If the selection period exceeds the period for the online data area, the system automatically selects the medium-term data area.

**Time from/to**

If you use the selection criterion *Time,* you select a time (from... to...) to specify the evaluated period of time (date from ... to ...). If the selection period exceeds the period for the online data area, the system automatically selects the medium-term data area.

If you change the selection, this selection applies for the duration of the session and is also used when you call the KPI monitor the next time.

The detail application *Failure report* cannot show any evaluations based on shifts. The selection fields *Filter mode time/shift* and *Shift* are therefore not available.

### *3.9.1.2 Field descriptions – List*

**KPI**

Numeric presentation of the KPI

**Machine**

Machine number and short description of the machine

**Full name of machine**

Full name of the machine

#### *3.9.1.3 Field descriptions – Detail*

**Machine**

Machine number and short description of the machine

**Graphic**

Graphic display of the relevant KPI in a diagram

# 4 Contacts

## 4.1 General

The application's aim is to search for contact persons within the company, to check if they are available and to contact them directly from the application (depending on the hardware in use).

## 4.2 Summary

The list of contacts shows information about the relevant persons and by clicking the phone or mobile number, the selected contact can directly be called. By clicking the e-mail address, a new e-mail opens to send a message to the contact. Both functions can only be used if the hardware in use supports this function.

A user name and password are required to log in to the "contact partners" application in the HR section (login via badge number and PIN code is not sufficient), as this function also shows data of other employees and not only data of the person logged in. The logged in user can only see the employees' data for whom they are authorized by responsibility areas.

The list of contacts shows the most important information about the persons, e.g.:

- Picture
- Company
- Area
- Cost center
- Function
- Phone numbers
- E-mail address

### Selection criteria

The search field of the header allows searching across all displayed values.

### Field descriptions - list

**Person's name**

The person's name showing the last name and first name

**Attendance**

Attendance statuses are represented by the following colors:

- Light-green: present
- Green: break

- Blue: planned absent
- Yellow: day off
- Red: unplanned absent
- White: person does not clock in/out
- Gray: no status

**Company phone**

The person's business phone number

**Mobile, company**

The person's business mobile number

**Company e-mail**

The person's business e-mail address

**Location**

Location where the person clocked in

**Planned present**

For absent employees this refers to the date when they are planned to be present

# 5 Messages Listing

## 5.1 General

The application provides an overview of the maintenance OPs and/or upcoming and thus active maintenance activities existing in the system. Consequently, the user is always aware of all maintenance activities.

### Selection criteria

The following selection criteria are available in the application:

**Resource**

Selects the specified resource.

**Resource type**

Selects the specified resource type.

**Classification**

Selects the classification of maintenances

### Field descriptions - list

**Resource**

Resource number.

**Resource designation**

Name of resource

**Resource type**

Type of resource

### Field descriptions - detail

#### "General" section

**Order**

This field is only relevant in connection with the additional feature "generate maintenance orders" or the "generation of calibration (inspection) orders. If this field is filled out the included order number refers to a maintenance/calibration order. The activity will automatically be reset if the maintenance/calibration order is finished. As the maintenance/calibration order is finished for this activity, the order number is also removed from this input field.

**Resource**

Shows the combined order and operation number.

**Resource designation**

Current status text of the operation.

**Activity**

Designation of the activity.

## "Information" section

**Information**

To ensure that the user or the maintenance worker receives more detailed information about running the activity (e.g. notes on regulations to be observed, materials to be used), a short description can be stored for each maintenance activity.

## "Assignment" section

**Project number**

This field is only relevant to the activity type "K" (calibration), whereas there are two different variants subject to system configuration.

Variant 1 (there is exactly one work plan for all calibration inspection plans):

=> Input of the calibration inspection plan number (without taking the version number into account)

Variant 2 (there is a separate work plan for each calibration inspection plan)

=> should remain empty.

.

**Planned order**

Control field that is currently not used. Consequently it remains empty.

**Cost object**

Control field that is currently not used. Consequently it remains empty.

**Activity type**

Identifies the activity type, calibrations, for example, are identified by the type K.

## "Resource information" section

**Inventory number**

Displays the inventory number defined in the resource configuration. Additional information including comments.

**Engraving number**

Shows the engraving number defined in the resource configuration. Additional information including comments.

**Drawing number**

Shows the drawing number defined in the resource configuration. Additional information including comments.

**Manufacturer**

Shows the manufacturer defined in the resource configuration. Additional information including comments.

**Owner**

Shows the owner name defined in the resource configuration. Additional information including comments.

### "Interval based on time" section

**Interval type**

Interval type for the activity. (Z = maintenance based on time, T = maintenance based on cycles, B = maintenance based on operating hours)

**Interval**

The period of time, after which the maintenance activity is to be run, should be entered here.

**Next activity**

Point in time when the next activity becomes due.

### "Interval based on operating hours" section

**Interval type**

Interval type for the activity. (Z = maintenance based on time, T = maintenance based on cycles, B = maintenance based on operating hours)

**Interval**

The period of time (number of operating hours) after which the maintenance activity is to be run, should be entered here.

**Hours recorded so far**

The time previously posted in HYDRA for this resource is shown here. This value is updated by a cyclical process. It is to be observed here that, for the previously recorded hours, only those RPA times are used, which have been marked as such in the resource type (option: *RPAs as hours of operation in the Maintenance Calendar*).

**Next activity**

Number of hours of operation after which the next activity becomes due.

### "Interval based on cycles" section

**Interval type**

Interval type for the activity. (Z = maintenance based on time, T = maintenance based on cycles, B = maintenance based on operating hours)

**Interval**

The number of machine cycles after which maintenance is to be carried out.

**Previously recorded cycles**

The number of resource cycles recorded so far in HYDRA is displayed here. This value is updated by a cyclic process.

**Next activity**

Number of cycles after which the next activity becomes due.

#### *5.1.1.1*